A.3: Warp the Images

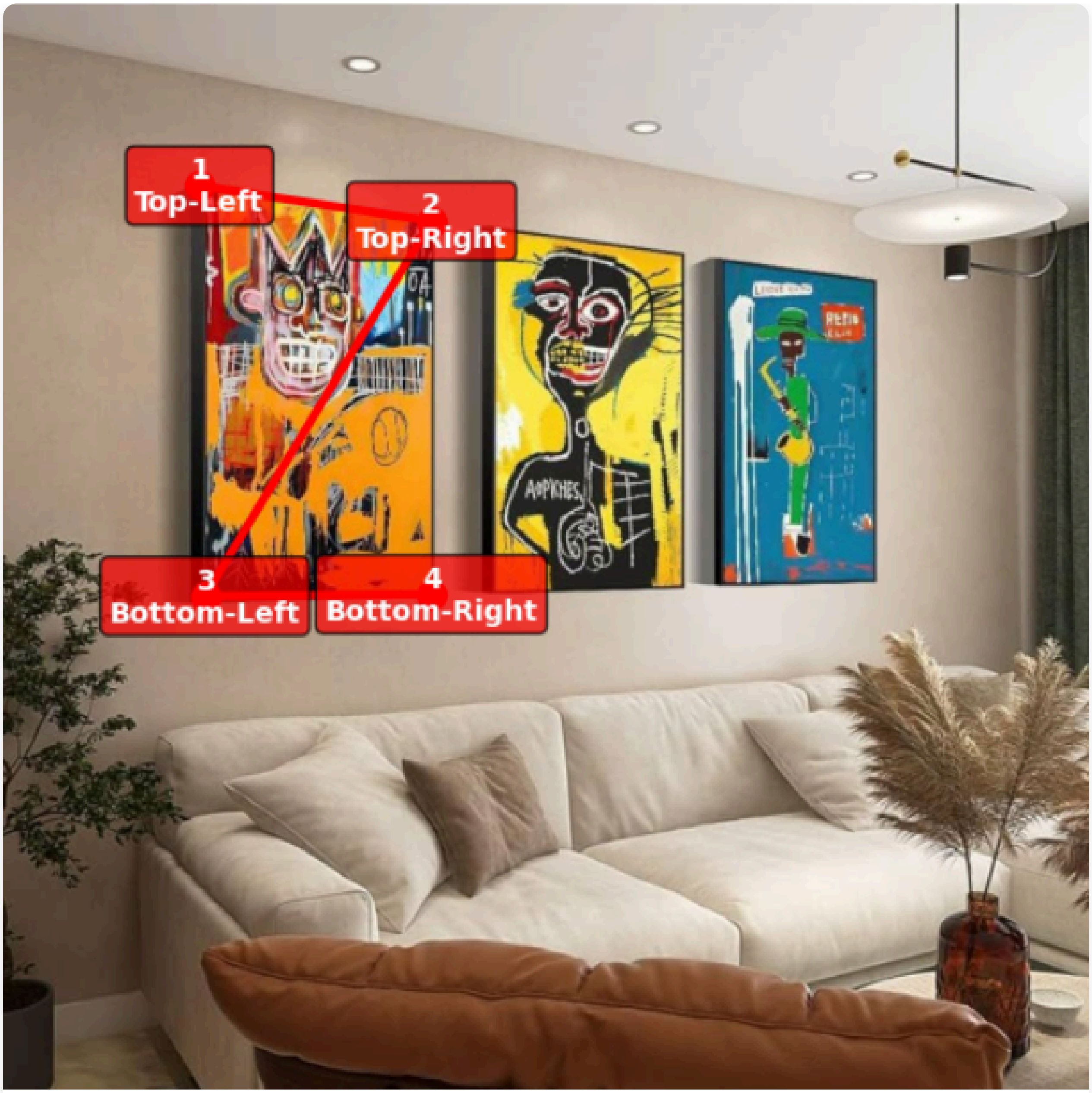
Implemented inverse warping with two interpolation methods: nearest neighbor and bilinear interpolation. Applied image rectification to demonstrate homography correctness.

Image Warping   **Rectification**

Applied homographies to rectify perspective-distorted rectangular objects back to their true shape, demonstrating the practical application of projective transformations.

Original with Quadrilateral

Rectified Result



Original with Quadrilateral

Rectified Region - Nearest Neighbor
(From Full Image)

Rectified Region - Bilinear
(From Full Image)

Rectified Result

Rectification Process: Selected four corners of a perspective-distorted rectangular object and computed a homography to map them to a perfect rectangle, effectively "undoing" the perspective distortion.

A.4: Blend Images into Mosaic

Blending Procedure

1. **1. Mosaic Bounds Computation**
Calculate final mosaic size by transforming all image corners through their homographies to determine the bounding box that contains all warped images.

2. **2. Three Blending Methods Implemented**

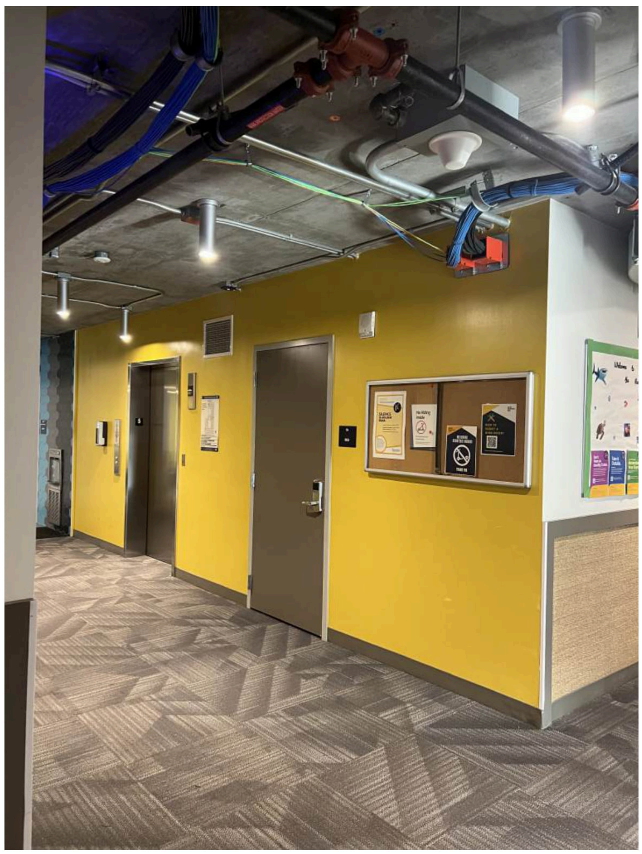
4

4. Results & Analysis

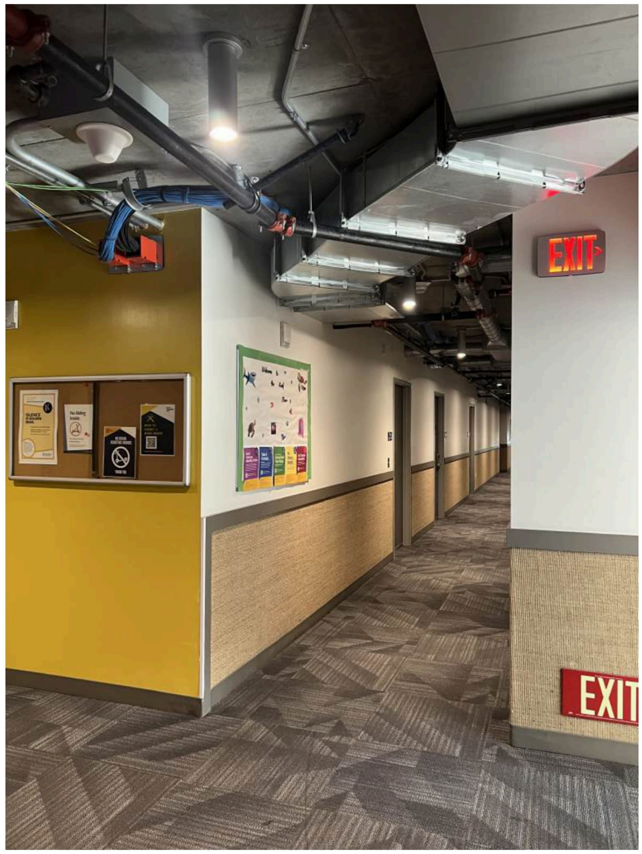
Created three mosaics with comparative analysis showing progressive quality improvement from simple → weighted → pyramid blending.

Created seamless panoramas by warping images into alignment and applying sophisticated blending techniques to eliminate visible seams and artifacts.

Mosaic 1 | Mosaic 2 | Mosaic 3 | Blending Methods



Source Image 1



Source Image 2

Final Mosaic

Conclusion

This project successfully implemented a complete image mosaicing pipeline from homography recovery to seamless blending. The results demonstrate professional-quality panorama creation with accurate geometric alignment and photometrically consistent blending.

Key Achievements:

- Accurate homography recovery with mean error of 4.80 pixels
- Implementation of two interpolation methods with performance analysis
- Successful rectification of perspective-distorted objects
- Creation of three high-quality mosaics using advanced blending techniques
- Comprehensive comparison of blending methods

CS 180 Project - Image Warping and Mosaicing
Implementation of homography estimation, image warping, and mosaic creation

4

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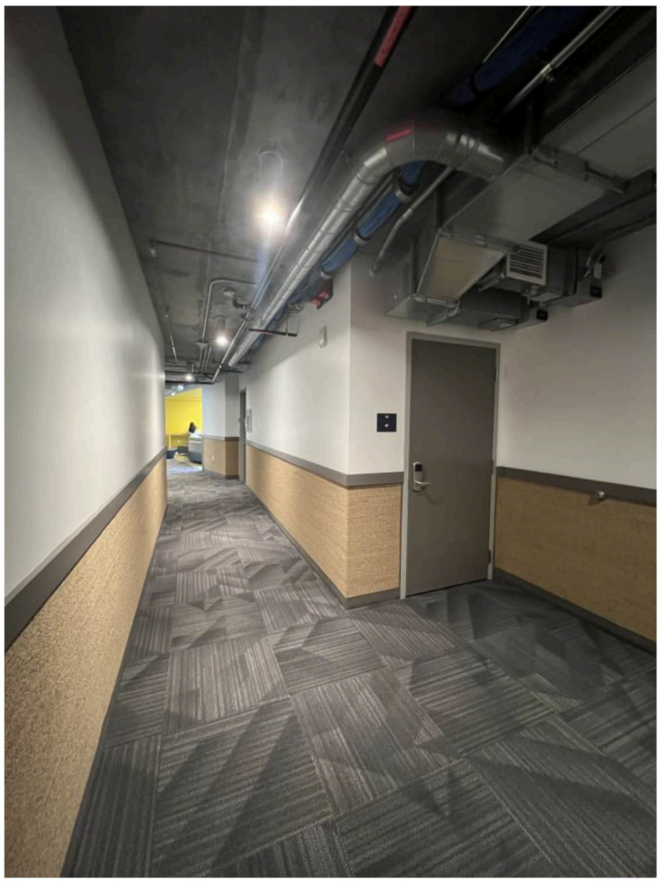
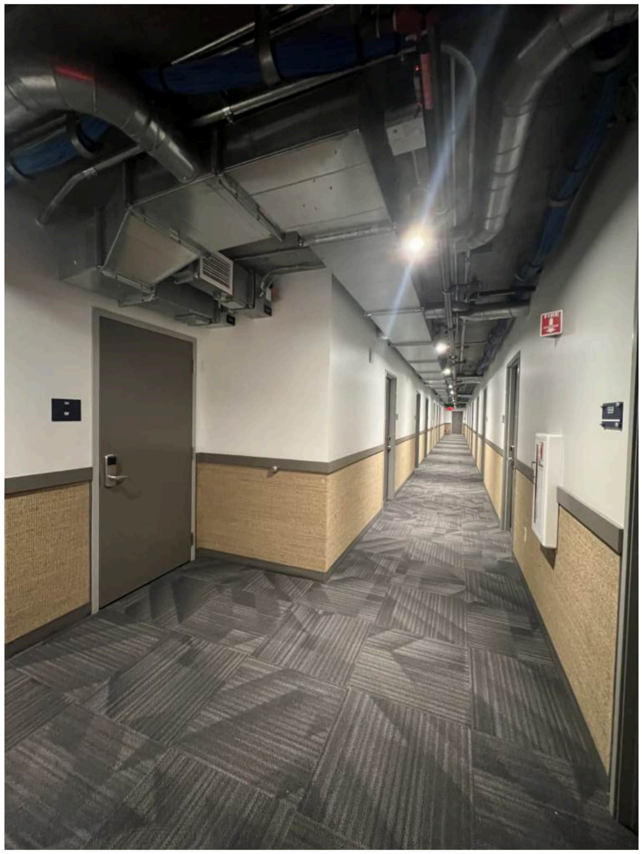
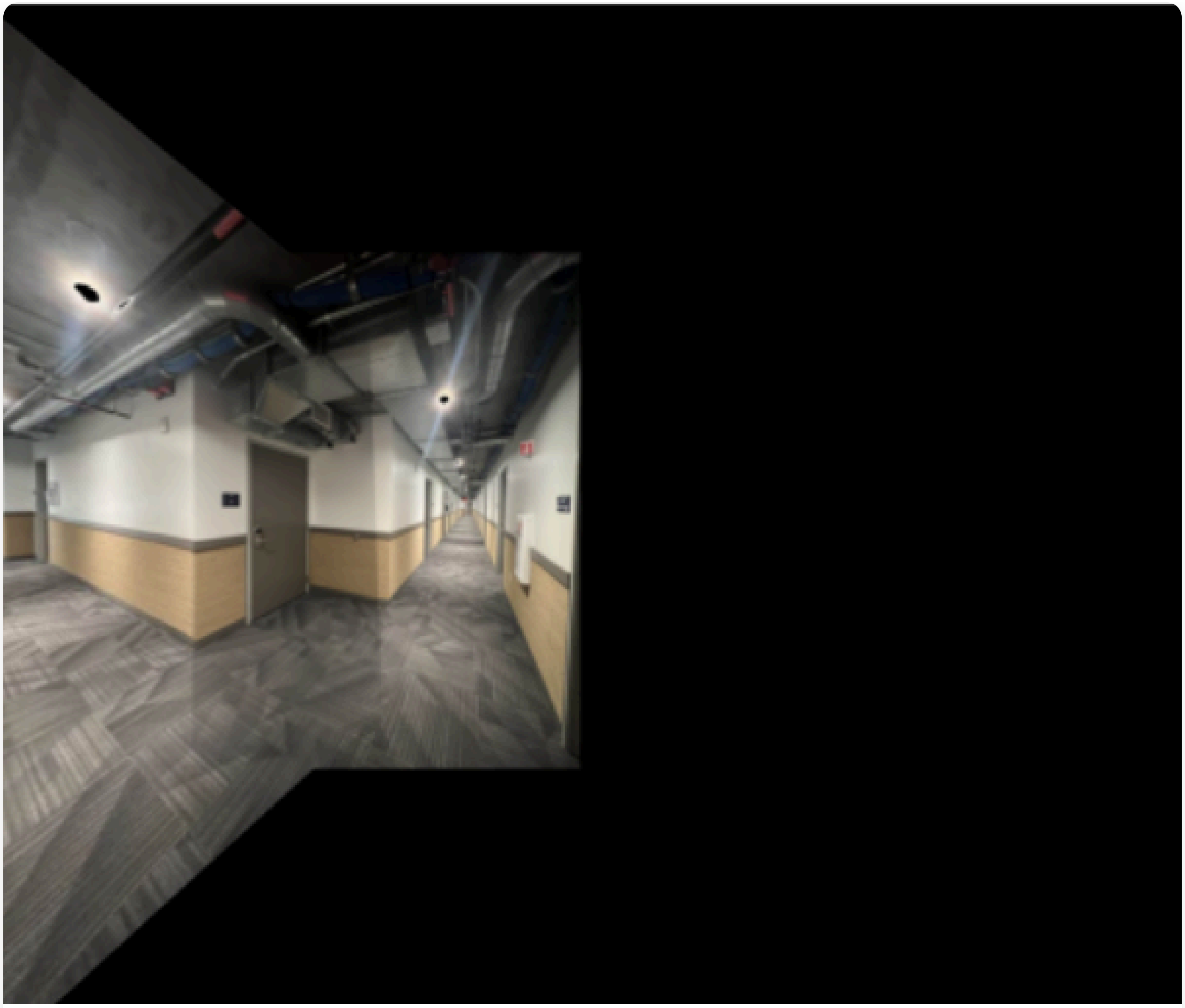
Mosaic 1 | Mosaic 2 | Mosaic 3 | Blending Methods



Source Image 1



Source Image 2



Final Mosaic

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Mosaic 1 Mosaic 2 Mosaic 3 Blending Methods

Source Image 1

Source Image 2

Final Mosaic

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Mosaic 1 | Mosaic 2 | Mosaic 3 | **Blending Methods**

Simple Overlay

Weighted Average

Laplacian Pyramid

Blending Analysis: The Laplacian pyramid approach produces the highest quality results by blending images at multiple frequency bands, preserving fine details while eliminating visible seams. Weighted averaging provides a good balance between quality and computational cost.

Conclusion

This project successfully implemented a complete image mosaicing pipeline from homography recovery to seamless blending. The results demonstrate professional-quality panorama creation with accurate geometric alignment and photometrically consistent blending.

Image Warping and Mosaicing

CS 180 Project - Creating seamless panoramas through homography estimation, image warping, and advanced blending techniques

Project Overview

This project implements a complete pipeline for creating image mosaics by recovering homographies between images, warping them into alignment, and blending them seamlessly. The implementation includes both geometric correction (rectification) and photometric blending for professional-quality results.

4.80 px
Mean Homography Error

3
Mosaics Created

2.4x
Bilinear vs NN Speed Ratio

3
Blending Methods

A.1: Shoot and Digitize Pictures

Captured photographs with a fixed center of projection and rotated camera to ensure projective transformations between images. Images were taken with significant overlap (40-70%) for reliable homography recovery.

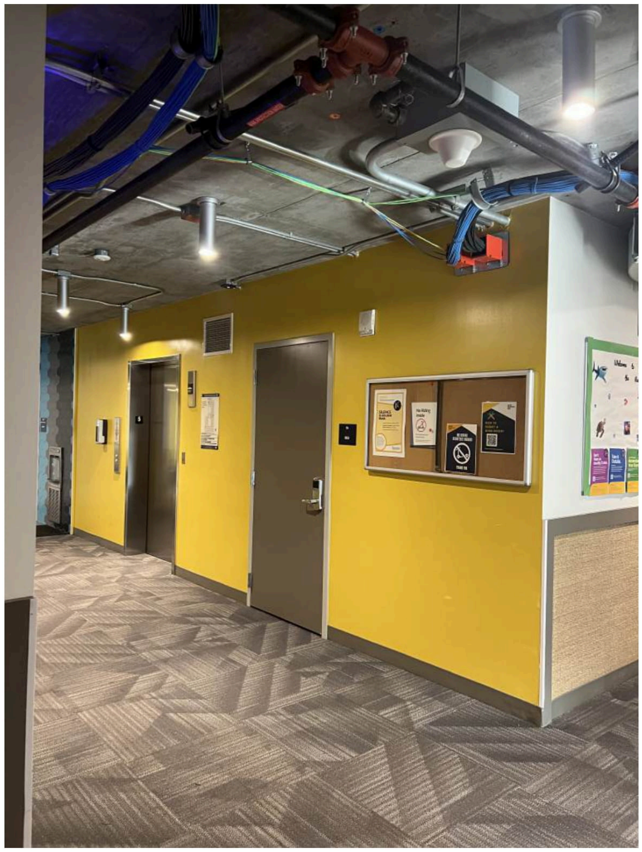

Image 1

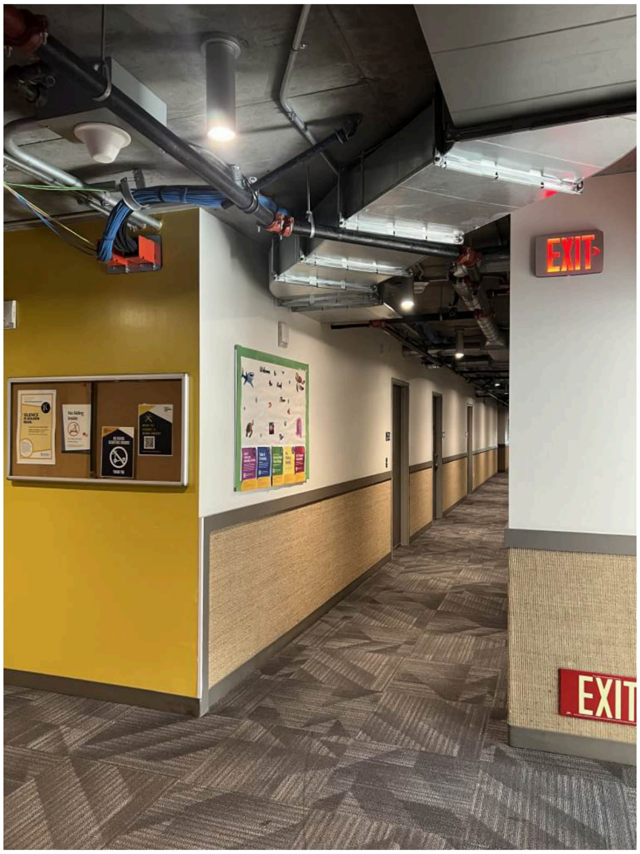

Image 2

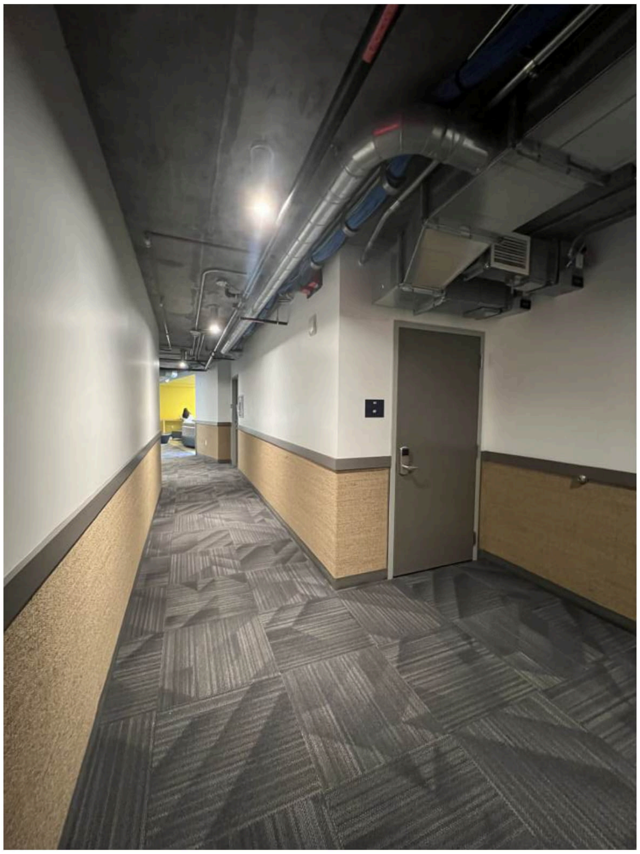

Image 3

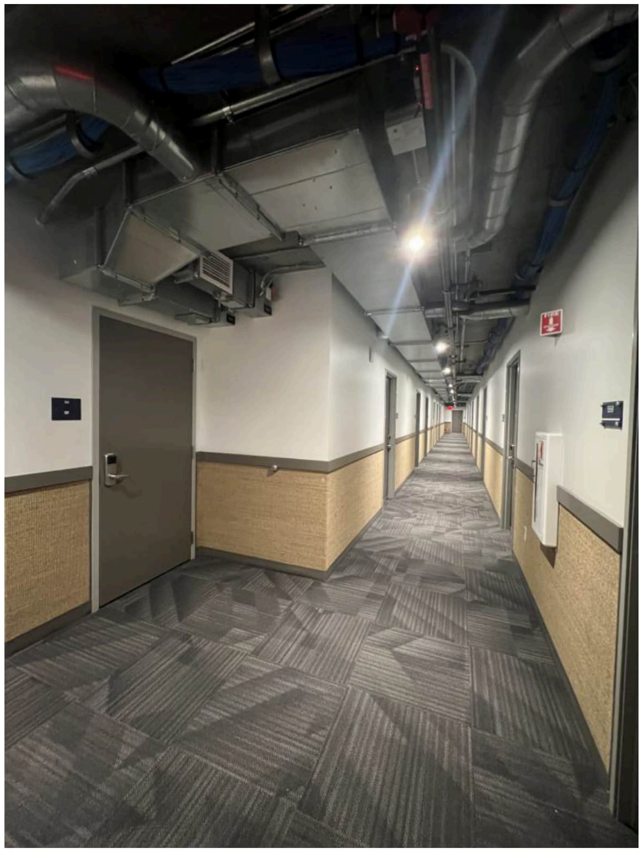

Image 4

Image 5

Image 6

Technical Details: Images were captured with careful attention to maintaining consistent lighting and avoiding lens distortion. The overlap ensures sufficient correspondences for robust homography estimation.

A.2: Recover Homographies

Implemented homography recovery using point correspondences and least squares estimation. The system computes a 3×3 transformation matrix that maps points from one image plane to another.

1. **Point Correspondence Selection**
   Manually selected 8 corresponding points between image pairs using an interactive interface.

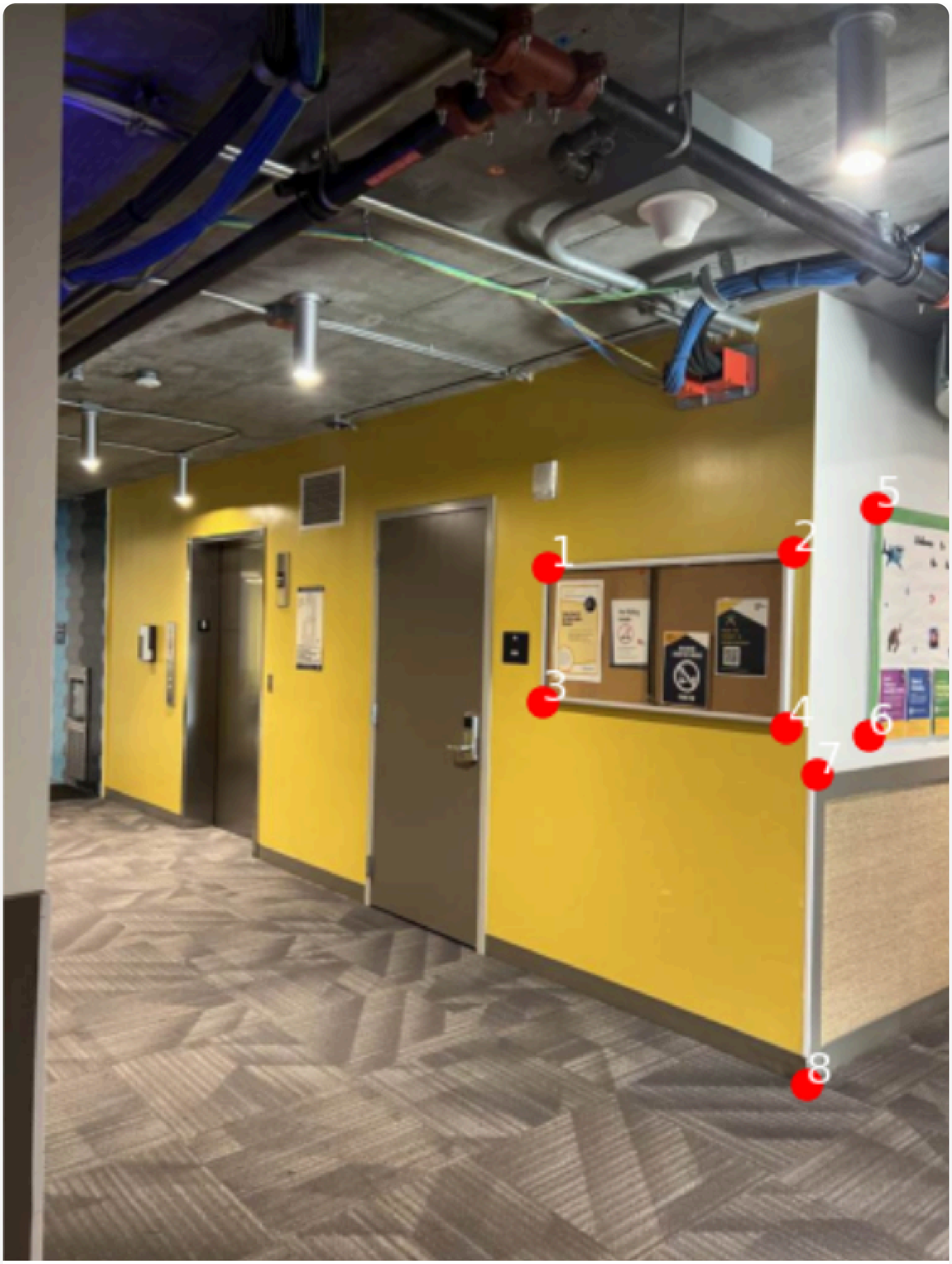

Image 1 with Points

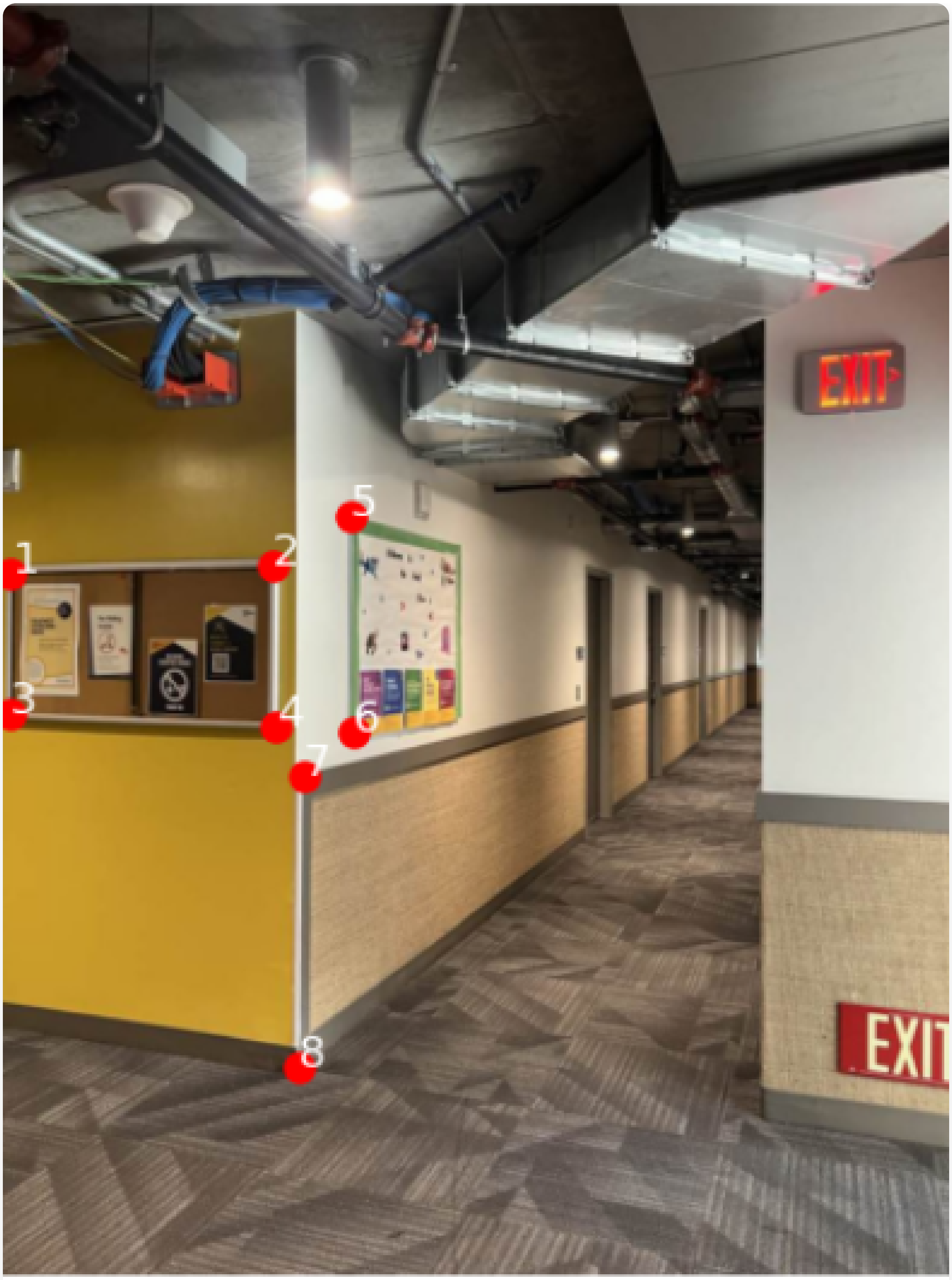

Image 2 with Points

2

Homography Computation

Solved the linear system Ah = b using least squares to recover the 8 degrees of freedom in the homography matrix.

```
# Homography Recovery Results
Rank of matrix A: 8/8
Condition number: 4.32e+08
Sum of squared residuals: 316.03

Recovered Homography Matrix H:
[[ 1.4043 0.0489 -1089.0651]
[ 0.1035 1.1970 -119.9642]
[ 0.0002 0.0000 1.0000]]
```

System of Equations: Ah = b

Where h = [a, b, c, d, e, f, g, h]^T

Matrix A (16×8) and vector b (16×1):

```
------------------------------
Eq 1: [ 1256.61 739.19 1 0 0 0 -744780.99 -438111.33] [x'1] =
592.69
Eq 2: [ 0 0 0 1256.61 739.19 1 -932917.41 -548781.04] [y'1] =
742.41

Eq 3: [ 1562.05 719.90 1 0 0 0 -1433049.55 -660449.33] [x'2] =
917.42
Eq 4: [ 0 0 0 1562.05 719.90 1 -1144608.99 -527515.77] [y'2] =
732.76

Eq 5: [ 1250.18 906.38 1 0 0 0 -740969.86 -537200.90] [x'3] =
592.69
Eq 6: [ 0 0 0 1250.18 906.38 1 -1145195.73 -830263.43] [y'3] =
916.02

Eq 7: [ 1552.40 938.53 1 0 0 0 -1429191.90 -864039.98] [x'4] =
920.63
Eq 8: [ 0 0 0 1552.40 938.53 1 -1446993.32 -874802.11] [y'4] =
932.10

Eq 9: [ 1664.93 665.24 1 0 0 0 -1688025.66 -674471.22] [x'5] =
1013.87
Eq 10: [ 0 0 0 1664.93 665.24 1 -1118292.12 -446827.24] [y'5] =
671.67
```

```
Eq 11: [ 1655.29 948.17 1 0 0 0 -1683568.46 -964374.67] [x'6] =
1017.09
Eq 12: [ 0 0 0 1655.29 948.17 1 -1553535.14 -889889.53] [y'6] =
938.53

Eq 13: [ 1590.98 996.40 1 0 0 0 -1520978.48 -952558.00] [x'7] =
956.00
Eq 14: [ 0 0 0 1590.98 996.40 1 -1580143.97 -989612.15] [y'7] =
993.19

Eq 15: [ 1578.12 1382.22 1 0 0 0 -1498536.17 -1312507.96] [x'8]
= 949.57
Eq 16: [ 0 0 0 1578.12 1382.22 1 -2135643.29 -1870524.63] [y'8]
= 1353.28
```

System Analysis: 16 equations for 8 unknowns (overdetermined system) solved using least squares with rank 8/8 and condition number 4.32e+08.

3

Accuracy Validation

Verified homography accuracy by measuring reprojection errors of the point correspondences.

4.80 px

Mean Error

7.85 px

Max Error

8

Points Used

A.3: Warp the Images

Implemented inverse warping with two interpolation methods: nearest neighbor and bilinear interpolation. Applied image rectification to demonstrate homography correctness.

Image Warping Rectification

Nearest Neighbor (0.575s)

Bilinear Interpolation (1.387s)

Performance Analysis: Bilinear interpolation produces smoother results but is approximately 2.4× slower than nearest neighbor due to the additional computations required for weighted averaging of four neighboring pixels.

A.4: Blend Images into Mosaic

Blending Procedure

1

1. Mosaic Bounds Computation

Calculate final mosaic size by transforming all image corners through their homographies to determine the bounding box that contains all warped images.

2

2. Three Blending Methods Implemented

Simple Overlay: Direct pixel replacement (fastest but shows edge artifacts)

Weighted Average: Distance-transform masks with smooth transitions

Laplacian Pyramid: Multi-resolution blending preserves details while eliminating seams

3

3. Technical Implementation

Inverse warping to avoid holes, alpha mask generation with border erosion, and optional center warp mode where both images transform toward common projection.

4

4. Results & Analysis

Created three mosaics with comparative analysis showing progressive quality improvement from simple → weighted → pyramid blending.

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Mosaic 1 Mosaic 2 Mosaic 3 Blending Methods

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Key Achievements:

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- Implementation of two interpolation methods with performance analysis
- Successful rectification of perspective-distorted objects
- Creation of three high-quality mosaics using advanced blending techniques
- Comprehensive comparison of blending methods